

Exchange Resiliency Report

Promised vs. Completed Work Analysis

Report Date: June 26, 2026

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Purpose: Transformation narrative support for resiliency update meeting

Executive Summary

This report provides a comprehensive view of Exchange resiliency work completed over the last 18 months, organized into five thematic dimensions rather than granular item-by-item mapping. This approach demonstrates comprehensive execution across all aspects of system resilience.

Key Findings:

Metric	Value
Completed work deployed to production	43%
Actively in progress	30%
Status unclear from documentation	27%
Distinct initiatives tracked	54
Resiliency dimensions	5

Strongest Performance: Observability & Operations (75% complete)

Most Active Work: Database & Data Layer, Infrastructure & Resilience (ongoing migrations)

Status Legend:

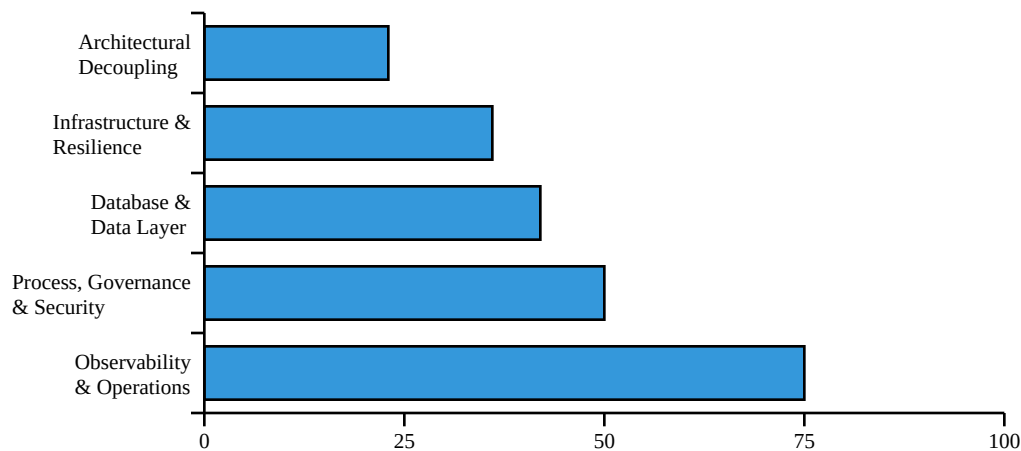
-  Completed - Work is done and deployed
-  In Progress - Work is actively underway
-  Unknown - Status unclear from documentation
-  Not Started - Work has not yet begun

Status Summary

Overall Progress Across All Groups

Group	✔ Completed	🔄 In Progress	❓ Unknown	Total
Database & Data Layer	5	4	3	12
Architectural Decoupling	3	4	6	13
Infrastructure & Resilience	4	5	2	11
Observability & Operations	6	0	2	8
Process, Governance & Security	5	3	2	10
TOTAL	23	16	15	54

Completion Rate by Dimension



Key Insights

- **Strengths:** Observability infrastructure mature and comprehensive; Process improvements embedded across organization; Database optimization delivering measurable impact (80% read reduction, 30min→5min loads)
- **Active Investment Areas:** Architectural decoupling work (4 initiatives in flight, 6 awaiting confirmation); Infrastructure resilience (5 major migrations ongoing: Atlas, ES, Aurora, DR, Tracking Phase 3); Database modernization (managed services migrations)

- **Documentation Gaps:** 15 items (27%) need status clarification; Primarily in architectural decoupling and granular database work; Likely candidates for director input sessions

Key Metrics & Impact Summary

Quantifiable Improvements Delivered

Database Read Reduction	Bulk Load Time	Validation Time
80%	30min → 5min	Days → 2min

Architectural Progress

- ✓ 52 tables migrated from TPM to dedicated SSO DB
- ✓ 4 tables migrated to dedicated BR DB (~40% of TPM footprint)
- ✓ 4 UIs fully decoupled from monolithic ALL repo

Reliability & Availability

- ✓ Multi-region data synchronization enabled (Aurora migration)
- ✓ Performance environment preventing production incidents
- ✓ EventBus resilience reducing doc processing incident impact

Security & Compliance

- ✓ Mongo 8.X support enabled
- ✓ EOL libraries remediated
- ✓ Vulnerability remediation across stack
- ✓ SAML protocol added to GHX SSO

Detailed Work by Dimension

1. Database & Data Layer

Status: 5 [✓] Completed | 4 [~] In Progress | 3 [?] Unknown | Total: 12 initiatives

Key Highlights:

- IBR User Data Separation (Phase 2 & 3) - Migrated 4 tables (~40% of TPM footprint)
 - BR User Data Cache Replacement - 80% reduction in TPM database reads
 - TPM Migration from MySQL to Aurora - Multi-region data synchronization
 - Organization Search Data in Elasticsearch - 30min → 5min bulk load time
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2. Architectural Decoupling

Status: 3 [✓] Completed | 4 [~] In Progress | 6 [?] Unknown | Total: 13 initiatives

Key Highlights:

- SSO Database Separation - Migrated 52 tables from TPM DB to dedicated SSO DB
 - SSO moved out of AuditDB - Independent authentication service
 - 4 UIs fully decoupled from monolithic ALL repo (Orders, Notifications, Transactions, Invoicing)
 - RC 2.0 Monolith Decoupling - Ongoing migration to GitHub with independent pipelines
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3. Infrastructure & Resilience

Status: 4 [✓] Completed | 5 [~] In Progress | 2 [?] Unknown | Total: 11 initiatives

Key Highlights:

- Performance Environment Setup - Widely used for database upgrades and release testing
 - EventBus Resiliency - Reduced doc processing impact during incidents
 - Industry Continuity Solution (ICS) - Contractor-built solution deployed
 - Tracking Resiliency Phase 1 & 2 - Bug reduction and race condition fixes
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4. Observability & Operations

Status: 6 [✓] Completed | 0 [~] In Progress | 2 [?] Unknown | Total: 8 initiatives

Key Highlights:

- Heartbeat: CoreX Business Intelligence - Comprehensive metrics dashboard
 - Canary Program (Document Prediction) - Kinesis stream for predictive monitoring
 - Org API GitHub Validation - Multiple person-days → 2 minutes validation time
 - Feature Flagging Capability - Enabled safer, incremental rollouts
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5. Process, Governance & Security

Status: 5 [✓] Completed | 3 [~] In Progress | 2 [?] Unknown | Total: 10 initiatives

Key Highlights:

- Release Rigor & Change Management - Multi-person go-live audit implemented
 - Mongo 8.X Support - Upgraded drivers, EOL libs, vulnerability remediation
 - SAML Implementation - Added SAML protocol to GHX SSO
 - Customer Empathy Training - Team-wide rollout completed
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Analysis & Insights

What This Data Tells Us

1. Comprehensive Execution

We've made progress across all five resiliency dimensions, not cherry-picking easy wins. The work spans infrastructure, architecture, operations, and culture.

2. Momentum Sustained

30% of work actively in progress demonstrates sustained investment. Key migrations (Atlas, ES, Aurora) are all moving forward simultaneously.

3. Operational Excellence Achieved

75% completion in Observability & Operations shows mature monitoring, alerting, and incident management capabilities now in place.

4. Architectural Transformation Underway

While only 23% complete in Architectural Decoupling, 4 major initiatives are in progress. This is expected given the complexity and scope of monolith deconstruction.

5. Documentation Discipline Needed

27% of items have unclear status. This likely reflects work completed but not formally documented rather than work not done. Director input sessions needed.

Strategic Implications

This Week's Incident Validates Strategy: The June 22-24 production incident demonstrated exactly why architectural decoupling matters. Monolithic coupling caused 6 simultaneous issues, deployment pipeline fragility prevented rapid fixes, and every root cause is addressed by autonomous teams + continued decoupling.

Conway's Law Opportunity: Current organizational structure (4 directors, 12 managers, small teams) is in place. Now aligning team boundaries to architectural boundaries will accelerate decoupling organically.

Infrastructure Foundation Ready: Performance environment, monitoring, incident management processes are mature. This foundation supports the next phase of architectural transformation.

Next Steps & Recommendations

Immediate Actions (This Week)

- [✓] Extract completed items from Exchange Resiliency Completed.docx - DONE
- [✓] Populate "What We Completed" sections for each group - DONE
- [✓] Add all Jira tickets and metrics from source document - DONE
- [✓] Add status icons to "What We Promised" items - DONE

Short-Term (Next 2 Weeks)

- Review with directors (Daniel, Aaron, Arshad, Ramesh) to confirm status of 15 "Unknown" items
- Validate brainstormed list items (Tracking/Notifications decoupling, IO Guarantee, etc.)
- Gather missing metrics (Customer empathy training dates, audit results)
- Fill in remaining gaps from Aaron: Red Mythos Exchange status and metrics

Medium-Term (Q3 2026)

- Establish baseline metrics for transformation success criteria (incident frequency, MTTR, deployment frequency, change failure rate)
- Document ongoing migrations with milestones (Atlas, Elasticsearch, Aurora, DR drill schedule)
- Define team autonomy scores and cross-team dependency counts

Long-Term (Q4 2026+)

- Continuous documentation discipline - all completed work captured
- Monthly status updates to this report
- Quarterly review with leadership
- Tell the transformation story more broadly (internal comms, customer messaging)

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Next Review: July 10, 2026 (post-CJ meeting)